ELM: linear SiPM and electronics clipping pilot

NAUSICAA

100 matched bb0nu events, IDs 0–99, for both gas models. E/P = 2 kV/cm/bar at 15 bar; 3 mm holes, 5 mm EL depth and 5 mm square-grid pitch. Electron collection is unity. VUV photons emitted uniformly on the hole axis are thinned through absorbing-hole acceptance, TPB conversion 0.65, forward fraction 0.5 and meridional capture $1 - 1/1.5$. The combined optical probability is 0.01386501. Guides have a 0.5 mm radial xenon gap; propagation and NOA 87 coupling are ideal. Saved photons are blue photons at the SiPM entrance; TPB is applied exactly once.

The 3×3 mm^2 S14160-3015PS uses packaged PDE 0.32 (peak-value approximation), gain 360000, and 700 kcps primary dark rate. Each optical/dark detection contributes full gain; microcell occupancy and recovery are deferred. The device has a silicone-resin entrance window; its transmission is included in PDE. The electronics adds 3.5417 PE RMS independent Gaussian noise and clips the 250 ns bucket-average current at 1 mA (4334.38 PE). Signed noise values are retained.

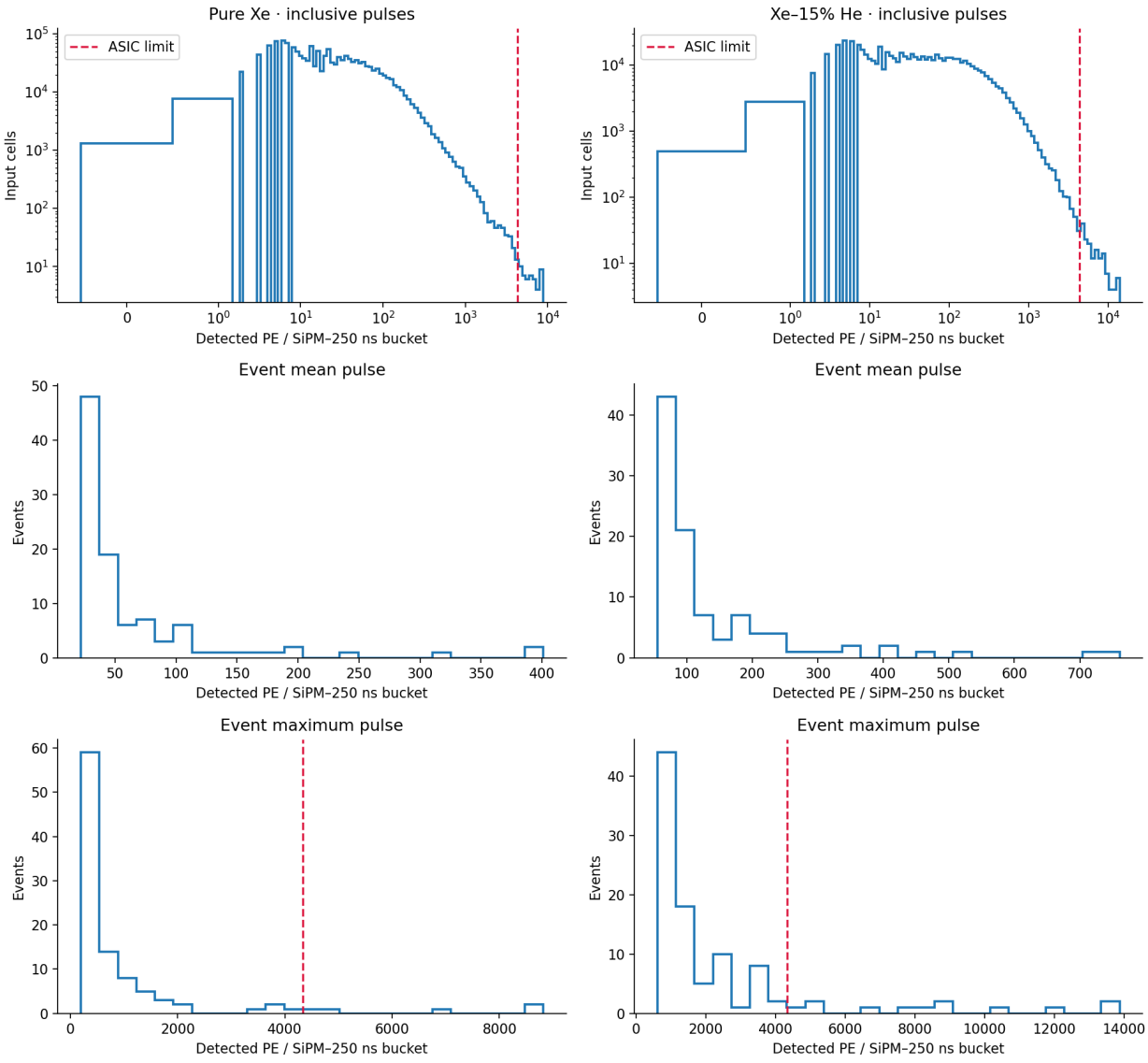
Gas	Events	Mean optical PE/event	Pulses clipped (%)	Events clipped (%)	Charge lost (%)	Largest pulse (PE)
Pure Xe	100	617,500	0.0037	5.0	0.1495	8,815
Xe–15% He	100	592,112	0.0270	11.0	0.6402	13,872

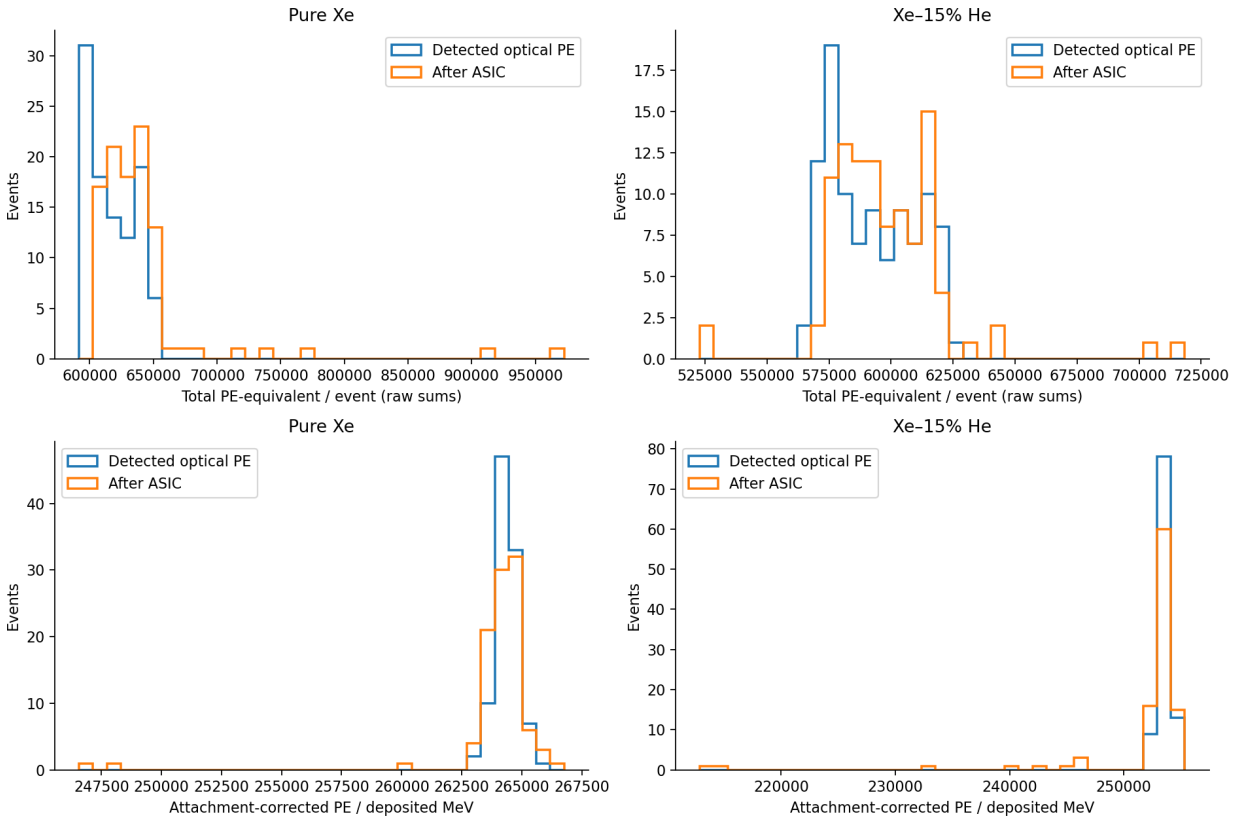
Gas	Inclusive mean	Inclusive median	Mean-pulse mean	Mean-pulse median	Max-pulse mean	Max-pulse median
Pure Xe	42.31	18.00	66.03	37.89	1,033.9	423.0
Xe–15% He	98.74	32.00	142.87	91.95	2,436.0	1,272.0

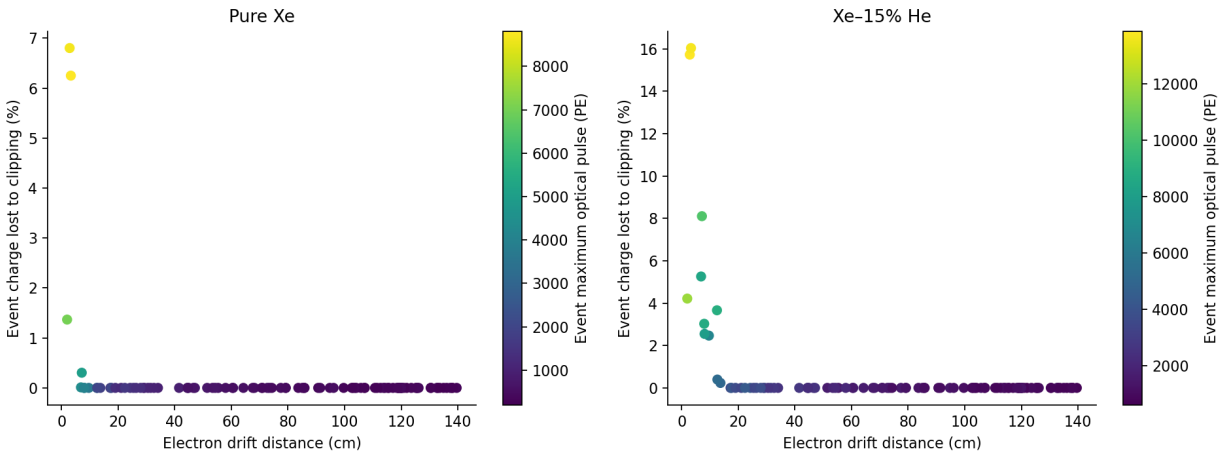
Event response and definitions

Gas	Detected PE RMS (%)	Before clipping RMS (%)	After clipping RMS (%)
Pure Xe	0.1803	0.2412	0.9411
Xe–15% He	0.1818	0.2351	2.4857

Pulse statistics select photon-positive SiPM–bucket cells and keep the same cells through electronics; zero detected PE in an input cell is retained. Each event contributes one mean and one maximum. Pulse clipping uses input cells as denominator; event clipping means any acquired bucket clips. Charge loss is summed clipped PE-equivalent divided by summed sensor PE-equivalent (optical plus dark). Event energy sums include all acquired cells on illuminated channels, over the event time span plus four quiet buckets at each end. Resolution diagnostics subtract the expected dark count, correct the known electron attachment survival and divide by deposited MC energy. Widths are full-distribution RMS/mean, including tails; they are not fitted peak widths. Before-clipping totals include electronics noise.


Pure Xe · inclusive pulses
ASIC limit
Input cells
Detected PE / SiPM–250 ns bucket
Xe–15% He · inclusive pulses
ASIC limit
Input cells
Detected PE / SiPM–250 ns bucket
Event mean pulse
Events
Detected PE / SiPM–250 ns bucket
Event mean pulse
Events
Detected PE / SiPM–250 ns bucket
Event maximum pulse
Events
Detected PE / SiPM–250 ns bucket
Event maximum pulse
Events
Detected PE / SiPM–250 ns bucket


Pure Xe
Detected optical PE
After ASIC
Events
Total PE-equivalent / event (raw sums)
Xe–15% He
Detected optical PE
After ASIC
Events
Total PE-equivalent / event (raw sums)
Pure Xe
Detected optical PE
After ASIC
Events
Attachment-corrected PE / deposited MeV
Xe–15% He
Detected optical PE
After ASIC
Events
Attachment-corrected PE / deposited MeV


Pure Xe
Event charge lost to clipping (%)
Electron drift distance (cm)
Event maximum optical pulse (PE)
Xe–15% He
Event charge lost to clipping (%)
Electron drift distance (cm)
Event maximum optical pulse (PE)

Largest event losses

Gas	Event ID	Electron drift (cm)	Maximum pulse (PE)	Charge lost (%)
Pure Xe	8	2.90	8,654	6.802
Pure Xe	49	3.32	8,815	6.250
Pure Xe	43	2.00	7,021	1.367
Pure Xe	66	7.18	4,992	0.303
Pure Xe	37	6.89	4,391	0.010
Xe–15% He	49	3.32	13,697	16.043
Xe–15% He	8	2.90	13,872	15.732
Xe–15% He	66	7.18	10,217	8.105
Xe–15% He	37	6.89	8,467	5.259
Xe–15% He	43	2.00	11,900	4.216

Validation and timing

Gas	Pair checks	Ledger checks	Measured PDE	PDE z-score	Quiet mean (PE)	Quiet RMS (PE)
Pure Xe	100	100	0.319975	-0.74	-0.0005	3.5412
Xe–15% He	100	100	0.320049	1.42	0.0027	3.5404

Photon replay/archive: 4.729 s; SiPM + electronics: 11.873 s; combined timed workflow: 16.602 s. These timings include first-use compilation within the stages, source checks and HDF5 I/O; process startup and presentation are separate.

The 22 dedicated optical/linear-response tests passed. Saved-event checks verify grid geometry, primary-pair identity, charge ledgers, PDE sampling and quiet-cell electronics noise.

Reproduction

Numerical source: `results/elm_pilot/elm_pilot.h5`.
File index: `results/elm_pilot/files.toml`.
Complete workflow: `make elm-pilot`.
Presentation only: `make elm-report`.

Input cards

```
schema = "nausicaa.electron_detector_spec.v1"
pressure_bar = 15.0
helium_fraction = 0.15
drift_speed_mm_us = 1.0
lifetime_ms = 15.0
hole_pitch_mm = 5.0
hole_diameter_mm = 3.0
hole_depth_mm = 5.0
sipm_shape = "square"
sipm_side_mm = 3.0
bucket_width_us = 0.25
reduced_field_kV_cm_bar = 2.0
j_variance_ratio = 0.030
j_status = "user_specified"
j_reference = "Oliveira Figure 6; corrected approximate user reading, 2026-11-04"

optical_model = "elm_blue_meridional"
electron_collection_efficiency = 1.0
```

```
[optics]
photon_band = "blue_tpb"
tpb_conversion_efficiency = 0.65
forward_fraction = 0.5
guide_refractive_index = 1.5
gap_refractive_index = 1.0
guide_length_mm = 100.0
gas_gap_radial_mm = 0.5
propagation_transmission = 1.0
coupling_transmission = 1.0
electrode_transmission = 1.0
coupling_adhesive = "Norland NOA 87"
guide_length_policy = "fixed_grading_field"
grading_field_kV_cm = 4.65
cathode_potential_V = 1200.0
entry_potential_V = 31500.0
exit_potential_V = 46500.0
sensor_plane_potential_V = 0.0
reference = "docs/elm.tex; isotropic axial EL, absorbing walls, meridional isotropic-in-core
↪  capture"

[pure_xe]
d_T_mm_sqrtbar_per_sqrtcm = 3.5
d_L_mm_sqrtbar_per_sqrtcm = 0.9
yield_gain_photons_per_electron_kV = 137.0
yield_threshold_kV_cm_bar = 0.73

[xe_he15]
d_T_mm_sqrtbar_per_sqrtcm = 1.6
d_L_mm_sqrtbar_per_sqrtcm = 0.75
yield_gain_photons_per_electron_kV = 139.0
yield_threshold_kV_cm_bar = 0.80

schema = "nausicaa.sipm_spec.v1"
response_model = "linear"
input_photon_band = "blue_tpb"
shape = "square"
side_m = 0.003
pde = 0.32
gain = 360000.0
dark_reference_rate_Hz = 700000.0
dark_reference_area_m2 = 9e-6
crosstalk = false
afterpulsing = false
gain_spread = false
seed = 2026

[acquisition]
channels = "illuminated_in_event"
additional_channels = []
quiet_before_buckets = 4
quiet_after_buckets = 4
empty_first_bucket = 0
empty_last_bucket = 7

[provenance]
device = "Hamamatsu S14160-3015PS, 3x3 mm2; nominal +4 V, 25 C"
pde = "0.32 peak PDE at 460 nm used as provisional TPB-spectrum average; packaged PDE includes
↪  entrance window"
gain = "3.6e5 nominal electrons per primary avalanche; full gain linear approximation"
dark_rate = "S14160-3015PS nominal primary dark rate 700 kcps at 25 C and +4 V"
window = "S14160-3015PS silicone resin, nominal n=1.57; unit external coupling used in pilot"
datasheet = "papers/SiPM/Hamamatsu_S14160_small_cells.pdf"

baseline_pe = 0.0
```

```
bucket_width_s = 2.5e-7
current_limit_A = 0.001
lower_clipping = false
noise = "independent_gaussian_pe"
noise_rms_pe = 3.5416666666666665
schema = "nausicaa.sipm_electronics_spec.v1"
seed = 2027
transfer = "bucket_average_current"

[provenance]
noise = "Same input charge RMS as 0.5 PE at gain 2.55e6; rescaled to gain 3.6e5"
range = "User design input: upper clipping at 1 mA; supersedes the provisional 300 PE limit"
timing = "User 0.25 us buckets; integrated charge divided by bucket duration"
```